

LCA of impossible - Myths & Legends

TALOS

THREE CIRCUITS A DAY. NO PLACE TO HIDE.



STEP ONE

Function before footprint

FUNCTIONAL UNIT

Manufacture and 30 years of operation of one 25 m bronze Talos providing continuous coastal defence for Crete. Operation includes three 1,046 km patrol circuits per day, one 2 t boulder throw per circuit and one red-heat deterrence event per month.

SERVICE LIFE

30 years

Explicit scenario

PATROL

3 circuits/day

1,046 km each

DEFENCE

Rock + heat

Separate use modules

THREE CALCULATION MODULES

01 - MANUFACTURE

Bronze shell, internal structure allowance, casting yield and finishing electricity.

02 - PATROL

Legged-locomotion energy over the full coastal distance using an explicit cost of transport.

03 - DEFENCE

Monthly heating plus the kinetic energy and aggregate mass of repeated boulder throws.

The functional unit is operational by design. A product-only footprint would describe the bronze statue; the patrol requirement tests the system that makes it Talos.

STEP ONE

What the sources require

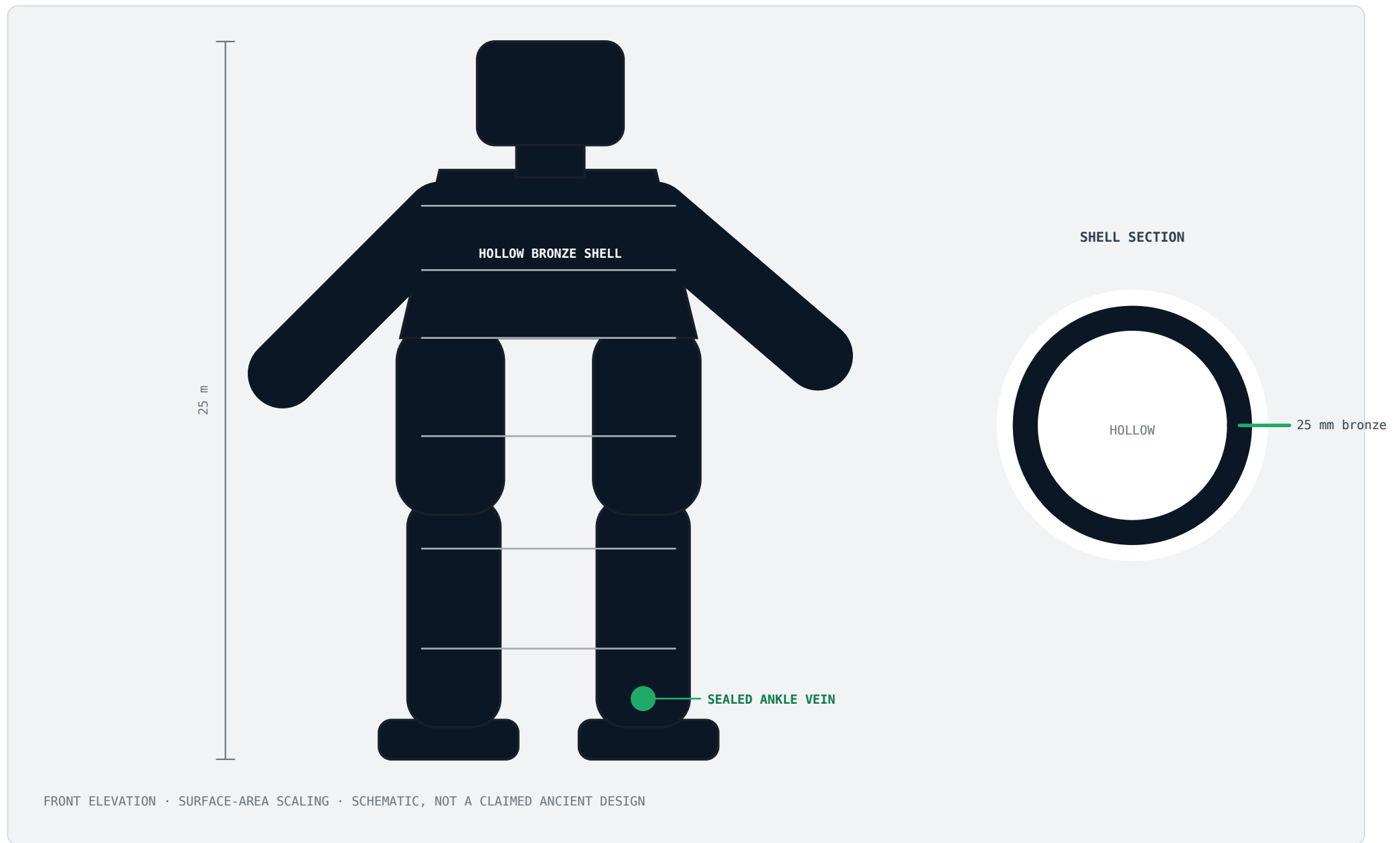
SOURCE REQUIREMENT	MODEL RESPONSE	STATUS
A bronze guardian of Crete	Hollow 25 m bronze shell with internal structure allowance	Modeled
Three circuits of the island each day	Three modern-coastline benchmark circuits per day	Modeled
Prevents ships landing by throwing rocks	One 2 t boulder throw per circuit at 50 m/s	Scenario
One vein from neck to ankle, sealed at the ankle	Single-point vulnerability shown; ichor excluded	Represented
Kills by a heated embrace	One monthly 20 to 700 deg C thermal event	Scenario
Height, shell thickness, power source and lifetime	Not provided in the ancient texts	Assumed

EVIDENCE RULE

The texts define the guardian and its behaviours. Geometry, route length, speeds, efficiencies and event frequency are modern reconstruction assumptions and remain editable in the workbook.

STEP TWO

A 98.1 t bronze machine



SHELL AREA

387.8 m²

Human area scaled by height

BRONZE SHELL

85.3 t

25 mm nominal thickness

FINISHED TALOS

98.1 t

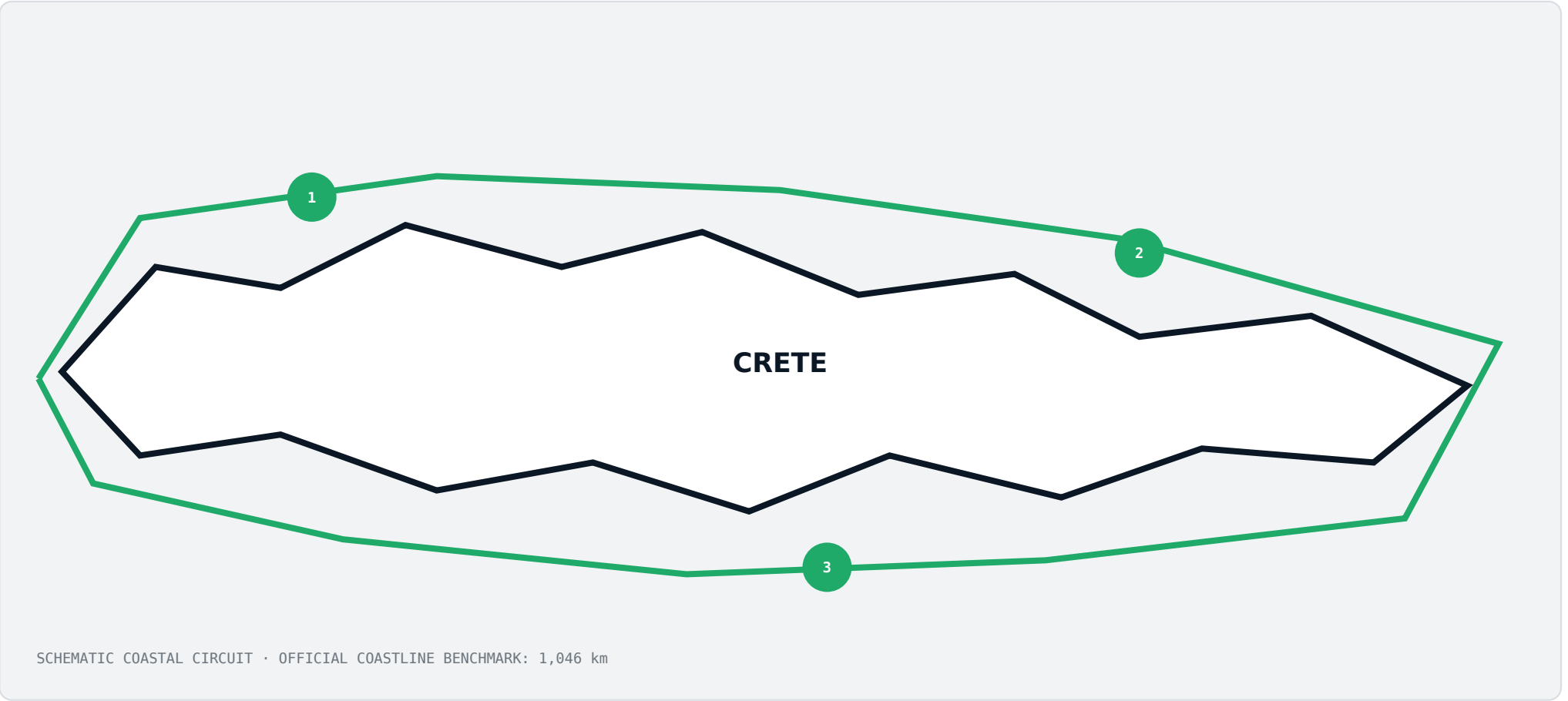
+15% internal structure

MANUFACTURING ASSUMPTIONS

- Surface area is scaled from a 1.75 m, 1.9 m² human reference: area grows with height squared.
- A 90% casting yield requires 109.0 t of primary metal input for the 98.1 t finished machine.
- Casting and finishing electricity: 87.2 MWh at 800 kWh per tonne of input.

STEP TWO

Three circuits every day



PER CIRCUIT

1,046 km

Modern benchmark

PER DAY

3,138 km

Three circuits

AVERAGE SPEED

130.8 km/h

One circuit every 8 h

THIRTY-YEAR DUTY CYCLE

OPERATING RHYTHM

00:00 -> circuit 1
08:00 -> circuit 2
16:00 -> circuit 3

LIFETIME SERVICE

32,850 circuits
34,361,100 km
Continuous coastal coverage

This is the decisive translation from myth to engineering: a narrative frequency becomes an extreme mobility requirement.

Route benchmark: Region of Crete official profile, 1,046 km of coastline. Modern coastline length is resolution-dependent and not an ancient route measurement.

STEP TWO

Patrol energy dominates

ENERGY MODEL

E = COT x m x g x d / efficiency

TERM	VALUE	ROLE
Cost of transport	0.20	Energy per unit weight-distance
Finished mass	98,102 kg	Moving machine mass
Lifetime distance	34.361 billion m	Three circuits/day for 30 years
Drivetrain efficiency	75%	Input-electricity conversion

LIFETIME ENERGY

2.4495 TWh

81.65 GWh per year

PATROL GWP

451.593 kt

DEFRA electricity proxy

TOTAL SHARE

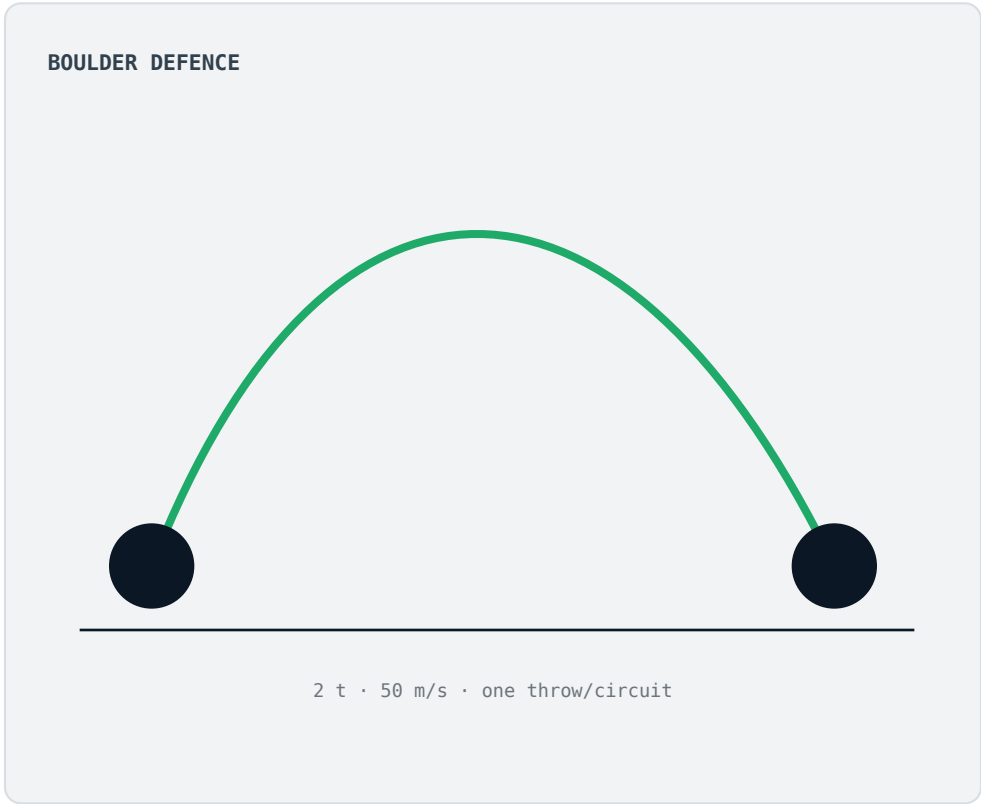
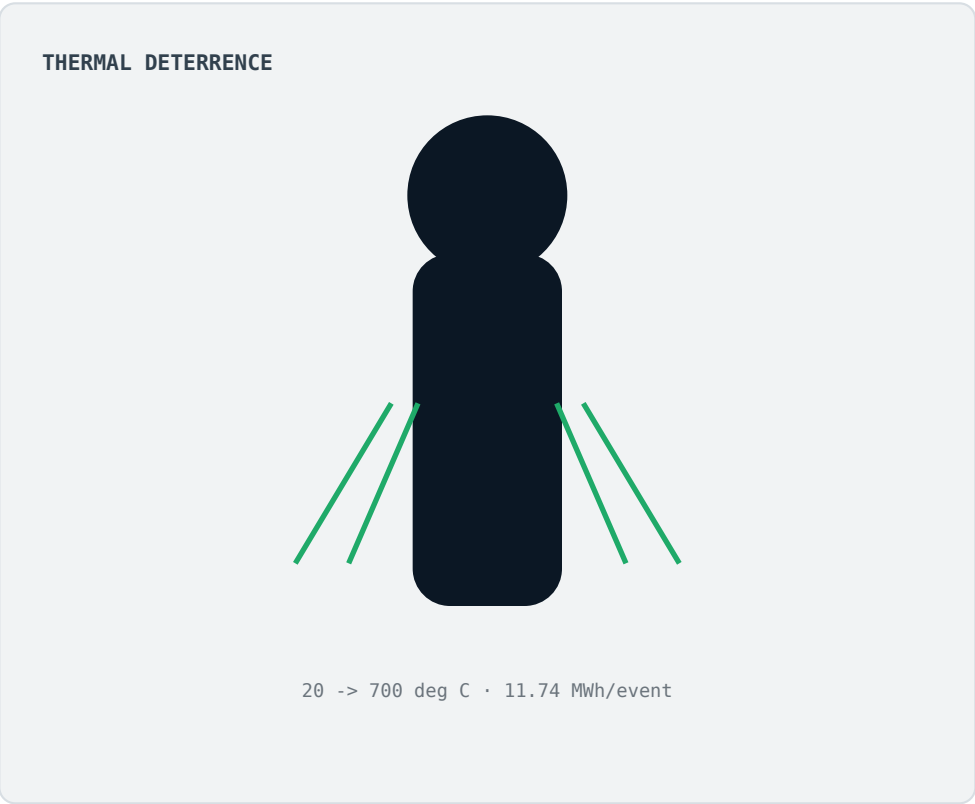
99.62%

System hotspot

The result scales linearly with COT, mass, distance and the inverse of efficiency. Every term is visible, editable and stress-tested.

STEP TWO

Two defensive subsystems



HEAT PER EVENT

11.74 MWh

98.1 t from 20 to 700 C

HEAT LIFETIME

4.225 GWh

360 monthly events

THERMAL GWP

778.9 t

Electricity equivalent

PROJECTILE INVENTORY

PARAMETER	LIFETIME VALUE	GWP
Boulder count	32,850 throws	-
Aggregate mass	65,700 t	512.7 t C02e
Launch electricity	38.0 MWh	7.0 t C02e

DEFENCE MODULE TOTAL

Thermal deterrence + boulder material + launch energy = 1.299 kt C02e.

Thermal and projectile modules are engineering scenarios anchored to ancient behaviours; frequency, temperature, rock mass and launch speed are assumptions.

STEP THREE

A modern proxy stack

FLOW / COMPONENT	FACTOR	APPLICATION
Metals, primary material use	3,821.94858 kg CO2e/t	Bronze proxy
Aggregates, primary material use	7.80307 kg CO2e/t	Boulders
Electricity generation	0.13096 kg CO2e/kWh	All electrical modules
T&D + both WTT components	0.05340 kg CO2e/kWh	Network + upstream
Combined consumption proxy	0.18436 kg CO2e/kWh	Calculated total

EXCLUDED

INCLUDED

- Primary metal and casting electricity
 - Patrol electricity equivalent
 - Monthly heating events
 - Boulder material and launch energy
- Ichor and control systems
 - Specific transport and infrastructure
 - Maintenance and replacement
 - Recovery and end of life

REPRESENTATIVENESS

DEFRA 2026 is strong for factor transcription and weak for Bronze-Age Crete. Arithmetic precision is kept separate from source and proxy quality.

STEP FOUR

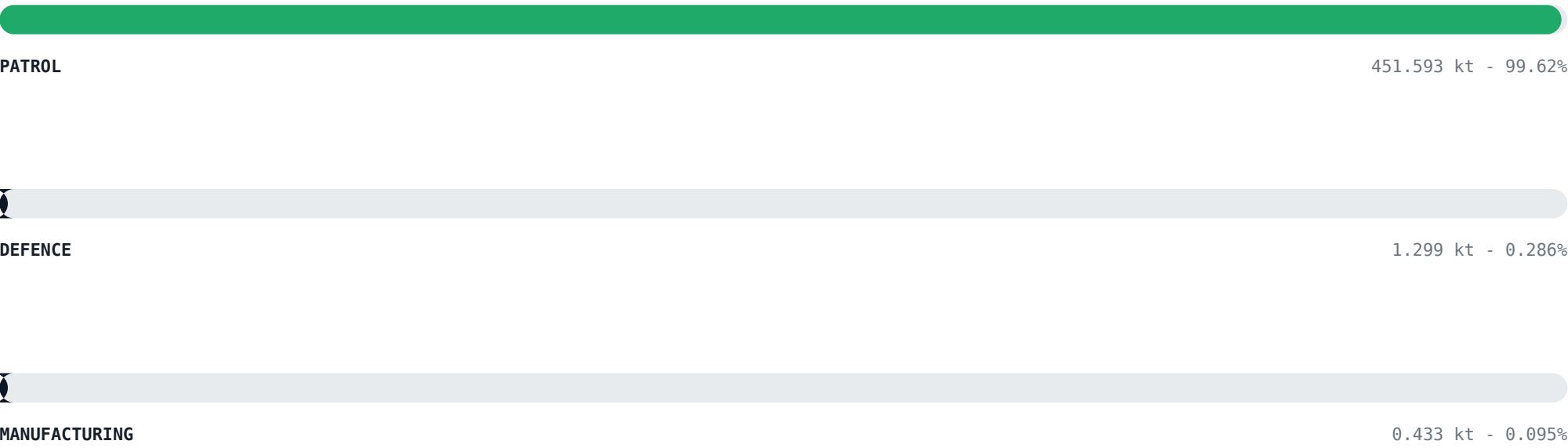
453.3 kt CO2e

MAIN RESULT

453.324 kt CO2e

13.80 t per circuit - 13.19 kg per km

CONTRIBUTION BY MODULE



HOTSPOT STATEMENT

The statue is not the system. The route is.

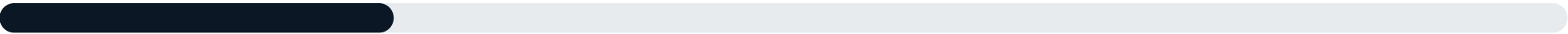
Manufacturing the bronze guardian is material-intensive, but it is numerically eclipsed by moving 98.1 tonnes around Crete more than thirty-two thousand times.

STEP FIVE

Mobility efficiency sets the result

CASE	COT	TOTAL	VS MAIN
Efficient	0.10	227.5 kt CO2e	-49.8%
Main	0.20	453.3 kt CO2e	baseline
Demanding	0.40	904.9 kt CO2e	+99.6%

TOTAL GWP



COT 0.10227.5 kt



COT 0.20453.3 kt



COT 0.40904.9 kt

DECISION CONSEQUENCE

Because the patrol term dominates, total GWP is almost proportional to COT. The locomotion architecture matters more than the bronze recipe.

STEP FIVE

Scale grows faster than height

HEIGHT	FINISHED MASS	TOTAL GWP	VS MAIN
15 m	35.3 t	163.5 kt C02e	-63.9%
25 m	98.1 t	453.3 kt C02e	baseline
35 m	192.3 t	888.0 kt C02e	+95.9%

15 M

35.3 t

163.5 kt C02e

25 M

98.1 t

453.3 kt C02e

35 M

192.3 t

888.0 kt C02e

SCALING RELATION

For constant proportions and shell thickness: area proportional to H2, mass proportional to H2, and patrol energy proportional to H2.

A 40% increase in height from 25 to 35 m nearly doubles the finished mass and the total modeled GWP. Mythic scale is a first-order design variable.

STEP SIX

Engineer the route first

FINDING 01

Manufacturing a 98.1 t bronze Talos contributes 0.433 kt CO2e under the selected modern proxies.

FINDING 02

The three-circuit duty cycle creates 34.36 million km of lifetime travel at a continuous average of 130.8 km/h.

FINDING 03

Patrol electricity contributes 451.593 kt CO2e and determines 99.62% of the modeled total.

DESIGN PRIORITY

- Reduce route distance, patrol frequency, mass or locomotion COT before optimizing bronze sourcing.
- Treat the 1,046 km modern coastline only as a transparent benchmark, not ancient cartographic truth.
- The sealed ankle remains a single-point failure that material mass does not solve.

Talos is a mobility system disguised as a statue.

LCA of the impossible - Season II

MYTHS & LEGENDS

**Where ancient stories meet
measurable emissions**